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AI-enabled innovation and firm growth in the Asia–Pacific region: Multilevel and experimental evidence on regulatory climate

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ABSTRACT

The present research investigates how AI-enabled innovation translates into firm productivity and economic growth under varying regulatory conditions. The primary goal of this study is to explain when and how firms are able to convert AI-enabled innovation into measurable productivity gains and sustained growth, and under which regulatory environments these effects are strengthened or constrained. Drawing on institutional and contingency perspectives, two complementary studies are conducted to test a moderated mediation model. Study 1 employs multilevel survey data from 420 firms across multiple regions, analyzed using hierarchical linear modeling. Results suggest that AI-enabled innovation is associated with firm growth both directly and indirectly through productivity gains, with the strength of the innovation–productivity link contingent on the regional regulatory climate. Study 2 adopts an experimental vignette design with 360 managerial participants to provide causal evidence. Regulatory clarity and supportive enforcement are manipulated, and results indicate that clear and supportive environments are associated with higher levels of intended AI investment, productivity expectations, deployment speed, and growth projections. Mediation analyses confirm that productivity expectations transmit the effects of regulatory conditions to growth, and moderated mediation findings reveal that the productivity pathway is strongest when both clarity and supportiveness are present. Together, these studies provide convergent evidence that institutional environments may amplify the benefits of AI adoption, offering actionable implications for innovation and regulatory policy by showing that environmental policies emphasizing regulatory clarity, consistent enforcement, and experimentation-friendly frameworks are likely to play an important role in enabling firms to translate AI investments into productivity improvements and economic growth.

1. Introduction

Artificial intelligence (AI) is increasingly recognized as a transformative force reshaping industries, altering competitive dynamics, and influencing broader patterns of economic growth (Trabelsi, 2024). According to recent estimates, AI-driven technologies are expected to contribute trillions of dollars to global GDP within the next decade, highlighting their potential to fundamentally reconfigure how firms create value and sustain growth (Gondauri, 2025). Yet despite this promise, a persistent puzzle remains: while some firms successfully leverage AI to enhance productivity and stimulate growth, others struggle to realize measurable returns from similar investments. Recent empirical evidence suggests that AI adoption alone does not guarantee uniform economic benefits, highlighting substantial heterogeneity in firm-level outcomes (Babina et al., 2025; Frimpong, 2025). Understanding why the outcomes of AI-enabled innovation vary so widely is critical for both scholars and practitioners concerned with the intersection of technology adoption, firm performance, and institutional environments (see Figs. 1 and 2).

Prior research has established that innovation, broadly conceived, is a key driver of productivity improvements and economic growth (Albis Salas et al., 2023; Shen et al., 2025). Within this stream, AI-enabled innovation represents a particularly powerful mechanism because of its capacity to automate complex processes, enable predictive analytics, and generate new business models (Yi & Ayangbah, 2024). Empirical studies have begun to demonstrate positive associations between AI adoption and firm-level outcomes such as efficiency, decision quality, and competitive advantage (Giachino et al., 2024; Rammer et al., 2021;

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Wang et al., 2024). However, the evidence remains fragmented and often limited to single-level analyses. As a result, limited insight exists into how AI-enabled innovation translates into firm-level economic growth through internal performance mechanisms such as productivity gains (Babina et al., 2025; Filippucci et al., 2024).

Institutional theory suggests that the external environment—particularly regulatory frameworks—plays a critical role in shaping how organizations deploy new technologies (Rudko et al., 2025). Supportive regulatory climates reduce uncertainty, provide incentives, and create legitimacy for innovation, whereas ambiguous or punitive regulatory approaches may deter risk-taking and slow adoption (Fenwick et al., 2018). In the context of AI, regulatory conditions are especially salient because of widespread concerns about ethical use, data privacy, and systemic risks (Ridzuan et al., 2024). Firms must therefore not only develop internal capabilities to innovate with AI but also navigate external institutional constraints and supports that influence how innovation translates into performance outcomes. Recent policy-oriented studies emphasize that regulatory clarity and enforcement consistency are central to determining whether AI investments yield productivity improvements and growth (Erdmann & Toro-Dupouy, 2025; Frimpong, 2025). Despite the importance of these dynamics, limited empirical evidence exists on how regulatory climates interact with firm-level innovation to shape productivity gains and growth trajectories.

This research seeks to address these gaps by developing and testing a multilevel moderated mediation model of AI-enabled innovation and economic growth. Specifically, we examine whether productivity gains serve as a key mechanism linking AI adoption to firm growth, and whether these effects are contingent on the degree of regulatory support within a firm's regional context. In doing so, the study responds to calls for greater clarity on when AI-enabled innovation translates into sustained economic outcomes rather than isolated performance improvements (Babina et al., 2025). Building on contingency perspectives, we argue that while AI-enabled innovation is likely to enhance productivity and growth (Rudko et al., 2021), the magnitude of these effects depends significantly on the surrounding regulatory environment. Regions characterized by clear, supportive policies are expected to magnify the benefits of AI innovation, whereas ambiguous or punitive environments may dampen or delay them.

This research contributes to theory and practice in several ways. First, it advances our understanding of how AI-enabled innovation translates into economic outcomes by identifying productivity gains as a central mediating mechanism. Second, it extends institutional perspectives by demonstrating that regulatory climates condition the effectiveness of technological innovation, offering a nuanced view of the interplay between internal resources and external environments. Third, it acknowledges that translating AI investments into productivity gains requires firms to reconfigure internal processes and routines, providing a capability-based rationale for heterogeneous outcomes without expanding the theoretical scope at this stage (Babina et al., 2025; Cimino, Corvello, et al., 2025). Fourth, it contributes methodologically by combining multilevel field data with experimental evidence, thereby providing convergent validity and stronger causal claims. Finally, the findings carry important implications for policymakers seeking to design regulatory frameworks that encourage innovation while ensuring responsible AI use, as well as for managers who must align their innovation strategies with institutional contexts to maximize returns.

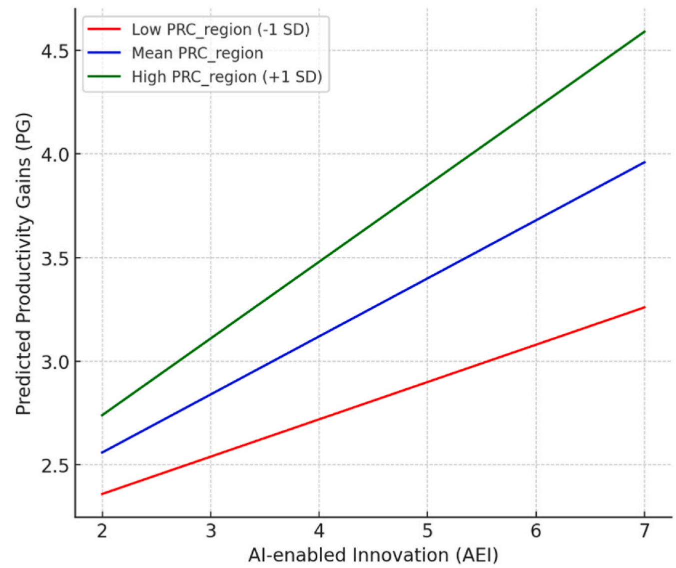

Fig. 2. Moderation effect.

In sum, this study addresses the pressing question of when and why AI-enabled innovation is associated with productivity improvements and economic growth. By bridging firm-level and institutional perspectives through a multi-method design, the research offers a comprehensive and contextually sensitive account of the mechanisms and boundary conditions that shape the economic value of AI adoption.

2. Theoretical framework and hypotheses development

2.1. AI-enabled innovation and productivity gains

AI-enabled innovation refers to the strategic integration of artificial intelligence technologies into a firm's products, services, and operations to generate value (Yi & Ayangbah, 2024). As AI increasingly reshapes organizational routines and workflows, firms that embed AI tools into their core functions tend to experience notable improvements in

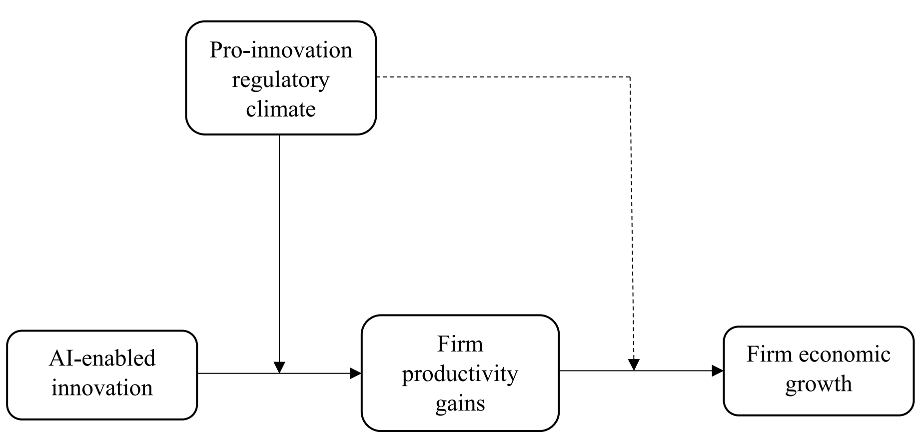

Fig. 1. Conceptual research model.

operational efficiency, process precision, and task automation (Li et al., 2021). These innovations facilitate the elimination of redundant steps, optimize data-driven decision-making, and enhance coordination across departments, resulting in better use of resources and reduced cycle times (Lin et al., 2024). In addition, AI-enabled innovation enables firms to automate knowledge-intensive tasks, uncover hidden operational bottlenecks, and personalize offerings at scale, which cumulatively boost productivity-related outcomes (Li et al., 2023). This is further supported by empirical findings indicating that firms adopting AI technologies outperform their peers in terms of production flexibility, service quality, and cost-efficiency (Larsen, 2021; Rudko et al., 2025; Wang et al., 2024).

Furthermore, from a capability-building perspective, AI-enabled innovation strengthens firms' dynamic capabilities by enabling them to sense, seize, and reconfigure resources in response to fast-changing environments (Sandeep et al., 2025). The ability to dynamically adapt processes and innovate using AI is directly linked to greater productivity through increased speed, lower cost, and improved output quality (Cui, 2025). AI systems, such as predictive analytics, machine learning models, and autonomous agents, can reduce human error and free up employee time for higher-order tasks, further amplifying productivity outcomes (Le & Behl, 2023). Prior empirical research has established that innovation is a strong predictor of productivity, especially when technological advancements are leveraged to redesign internal operations and service architectures (Alam et al., 2023; Damioli et al., 2002; Yi & Ayangbah, 2024). In line with the innovation-performance literature, firms that actively invest in AI to create new solutions tend to outperform those that rely solely on traditional innovation approaches (Brem et al., 2021). Thus, AI-enabled innovation is not only a symbolic investment but a performance-driving mechanism that enhances firm-level productivity across multiple dimensions. Accordingly, we hypothesize.

H1. AI-enabled innovation is positively associated with firm productivity gains.

2.2. Productivity gains and firm economic growth

Productivity gains represent a firm's enhanced ability to convert inputs into valuable outputs through improved efficiency, effectiveness, and resource optimization (Gao & Feng, 2023). As firms streamline operations, minimize redundancies, and optimize task execution, they unlock cost savings and accelerate throughput, which are critical precursors to revenue growth and market expansion (Zhai & Liu, 2023). These efficiency improvements allow firms to reallocate resources toward strategic initiatives such as market development and innovation, thereby supporting economic performance (Alam et al., 2023). In particular, productivity improvement enhances a firm's competitive positioning by lowering operational costs, increasing output volume, and improving service or product quality—all of which contribute to superior financial outcomes (Albis Salas et al., 2023). Empirical studies have consistently found that firms with higher productivity levels report faster sales growth, better profit margins, and increased market share over time (Gondauri, 2025). Moreover, higher productivity enhances firms' resilience by enabling faster adjustment and sustained profitability in volatile environments (Challoumis, 2024).

From a strategic growth perspective, productivity gains create a scalable foundation for long-term expansion. Firms that operate more efficiently are better equipped to enter new markets, diversify offerings, and invest in growth-oriented capabilities (Babina et al., 2025). Such firms also tend to attract greater external investment and stakeholder confidence due to their demonstrated ability to generate returns through operational excellence (Yang, 2022). In the context of AI-enabled innovation, productivity gains become even more critical, as AI-driven efficiencies translate into faster go-to-market cycles and improved agility—key growth accelerators in dynamic industries (Atienza-Barba et al., 2024). Accordingly, we hypothesize.

H2. Firm productivity gains are positively associated with firm economic growth.

2.3. Mediating role of productivity gains

While AI-enabled innovation directly enhances firm competitiveness, much of its impact on firm economic growth occurs through improvements in internal productivity. The strategic integration of AI technologies enhances operational efficiency, automation, and decision precision, thereby improving productivity through process reengineering, intelligent automation, and real-time data processing (Chin et al., 2025; Damioli et al., 2021). These improvements reduce inefficiencies and free up resources, enabling firms to pursue expansion strategies such as market penetration and innovation investment (Cimino et al., 2025a).

From a dynamic capabilities perspective, AI enables firms to reconfigure resources and adapt processes in response to environmental changes, with such transformations first reflected in productivity gains before translating into broader growth outcomes (Cui, 2025; Kumar et al., 2025; Renfei & Zhongwen, 2025). Empirical evidence further suggests that innovation contributes to economic performance primarily when firms successfully internalize efficiency gains and convert them into competitive advantage (Almheiri et al., 2024; Cimino et al., 2025b; Mele et al., 2024). Accordingly, productivity gains represent a key transmission mechanism linking AI-enabled innovation to firm economic growth. Accordingly, we hypothesize.

H3. Firm productivity gains mediate the positive relationship between AI-enabled innovation and firm economic growth.

3. The moderating role of pro-innovation regulatory climate

The effectiveness of AI-enabled innovation in generating productivity gains depends not only on internal capabilities but also on the external institutional environment. According to Institutional Theory, regulatory conditions shape how firms deploy and benefit from technological innovation (Rudko et al., 2025). A pro-innovation regulatory climate—characterized by supportive policies, incentives, and clear legal frameworks—reduces uncertainty and facilitates AI adoption (Li et al., 2021; Ridzuan et al., 2024).

Such environments provide firms with access to institutional support, including infrastructure, incentives, and collaborative platforms, which accelerate AI integration and enhance productivity outcomes (Fenwick et al., 2018; Renfei & Zhongwen, 2025). Conversely, ambiguous or restrictive regulatory conditions may hinder implementation and reduce the effectiveness of AI-enabled processes (Shankar & Gupta, 2024). Empirical evidence indicates that productivity gains from digital innovation are stronger in contexts with regulatory clarity and supportive policy regimes (Chan et al., 2024; Frimpong, 2025; Lin et al., 2024). Accordingly, regulatory climate acts as a boundary condition that strengthens the innovation–productivity relationship. Accordingly, we hypothesize.

H4. The positive relationship between AI-enabled innovation and productivity gains is strengthened when the pro-innovation regulatory climate is high.

4. Moderated mediation: the contingent path from AI-enabled innovation to firm growth

Building on the preceding arguments, the relationship between AI-enabled innovation and firm growth can be understood as a conditional indirect process. Specifically, the extent to which AI-enabled innovation translates into firm economic growth through productivity gains depends on the level of pro-innovation regulatory climate (Hayes, 2015; Preacher et al., 2007). In supportive regulatory environments, firms are better positioned to convert AI investments into productivity improvements that can be scaled into growth outcomes (Xia et al., 2025;

Eng'airo, 2024). In contrast, regulatory ambiguity and compliance burdens may weaken the productivity pathway, limiting the translation of innovation into economic performance (Al-Dulaimi & Mohammed, 2025). Prior research also suggests that favorable policy environments enhance firms' ability to leverage productivity gains for market expansion and sustained growth (Filippucci et al., 2024; Larsen, 2021; Yi & Ayangbah, 2024). Taken together, these arguments suggest that the indirect effect of AI-enabled innovation on firm growth via productivity gains is contingent on the regulatory environment.

H5. The indirect relationship between AI-enabled innovation and firm economic growth via productivity gains is stronger when the pro-innovation regulatory climate is high (i.e., moderated mediation).

5. Materials and methods

5.1. Sample and data

This research employed a mixed-method design consisting of a multilevel field survey (Study 1) and a complementary experimental vignette study (Study 2). The two studies were designed to address the same theoretical model using different empirical strategies: Study 1 provides externally valid evidence based on organizational data, while Study 2 strengthens internal validity and causal inference by experimentally manipulating regulatory conditions.

For Study 1, the target population consisted of firms operating across multiple regions with active engagement in technological adoption and digital transformation. This population was selected because such firms are at the forefront of leveraging artificial intelligence to enhance operational efficiency and economic outcomes (Cui, 2025; Zhai & Liu, 2023), thereby offering an appropriate context for examining the role of AI-enabled innovation. The firms included in Study 1 were drawn from multiple Asia-Pacific economies, including China, Malaysia, Singapore, and Thailand. While the sampling strategy ensured representation across regions and industries, the data were not intended to support country-specific comparisons. Instead, the focus was on capturing cross-regional variation in regulatory climate and its interaction with firm-level innovation.

A purposive sampling approach was employed to ensure that participating firms had already adopted or were in the process of adopting AI technologies. Within each firm, managerial respondents were approached, as they possess direct knowledge of innovation practices, productivity improvements, and growth performance. A stratified sampling approach across industries was used to capture variability in innovation practices while maintaining comparability. Questionnaires were distributed physically, with follow-up visits conducted to maximize response rates. A total of 420 useable responses were obtained, which is consistent with established guidelines for hierarchical linear modeling.

For Study 2, the target population comprised managerial decision-makers from firms engaged in or intending to engage in AI-related initiatives. Managers were selected because they shape AI-related resource allocation, deployment timelines, and compliance approaches. Participants were recruited through industry associations and executive education cohorts. A total of 360 managers participated and were randomly assigned to one of four experimental conditions (approximately 90 per cell) using a 2 × 2 between-subjects factorial design manipulating regulatory clarity (clear vs. ambiguous) and enforcement posture (supportive vs. punitive). All responses were collected in person using printed materials to avoid discussion effects.

Participants in Study 2 were recruited from managerial pools across similar Asia-Pacific contexts, although the experimental design did not explicitly model country-level differences, as the focus was on isolating causal mechanisms related to regulatory clarity and enforcement conditions.

Ethical procedures were consistently applied across both studies. Participants were informed about the academic purpose of the research, assured of confidentiality and anonymity, and informed of their right to withdraw at any time. No personally identifiable information was collected, and institutional approval was obtained prior to data collection.

5.2. Measures of variables

All constructs were measured using previously validated scales and adapted to the context of AI-enabled innovation and firm performance. Unless otherwise noted, Likert-scale items ranged from 1 (strongly disagree) to 5 (strongly agree) in Study 1, while numeric or 7-point scales were used in Study 2 where appropriate. Reliability and validity statistics for Study 1 are reported in Table 2.

AI-enabled innovation was measured in Study 1 using a four-item scale adapted from prior innovation research (Chen et al., 2014; Subramaniam & Youndt, 2005). Items captured the extent to which firms integrate artificial intelligence into product development, process improvement, and service delivery ($\alpha = .91$, CR = .93, AVE = .67). In Study 2, AI-enabled innovation was operationalized as intended AI investment, measured as the percentage of the IT budget managers intended to allocate to AI initiatives in the upcoming year.

Productivity gains were assessed in Study 1 using a three-item measure adapted from operational performance research (Ketokivi & Schroeder, 2004), capturing efficiency, output quality, and resource utilization ($\alpha = .88$, CR = .91, AVE = .65). In Study 2, productivity was captured through expected productivity gains, measured as managers' percentage-based expectations regarding operational improvement following AI deployment.

Firm economic growth in Study 1 was measured using a four-item perceptual scale adapted from Delmar et al. (2003) and Zahra and Covin (1995). The scale explicitly captured managers' assessments of growth in sales, market share, and profitability over the past three years relative to key competitors, demonstrating strong reliability ($\alpha = .85$, CR = .89, AVE = .62). In Study 2, firm growth was measured as expected growth, operationalized as managers' projected percentage growth in the coming year.

Pro-innovation regulatory climate was measured at the regional level in Study 1 using a four-item scale aggregated from managerial perceptions (Kostova, 1997; Xu & Shenkar, 2002). The scale assessed the extent to which local regulations, incentives, and policies support technological experimentation and AI adoption ($\alpha = .87$, CR = .90, AVE = .64). Interrater agreement and reliability statistics ($r_{wg} = .82$, ICC(1) = .18, ICC(2) = .73) supported aggregation. In Study 2, regulatory climate was experimentally manipulated through vignette scenarios varying regulatory clarity and enforcement posture, with manipulation checks confirming successful differentiation between conditions.

Control variables in Study 1 included firm size (log number of employees), firm age (years since founding), IT budget intensity (percentage of sales invested in IT/AI), and regional digital infrastructure (broadband penetration). In Study 2, a brief risk aversion scale and baseline AI maturity indicator were included for robustness checks; results were substantively unchanged, and parsimonious models are reported.

5.3. Models and data analysis procedure

In Study 1, hypotheses were tested using hierarchical linear modeling to account for the nested structure of firms within regions. Firm-level predictors were specified at Level 1, while the pro-innovation regulatory climate was modeled at Level 2. Prior to hypothesis testing, confirmatory factor analysis demonstrated clear discriminant validity among constructs, indicating that the measures did not collapse into a single latent factor. In addition, Harman's one-factor test revealed that the first unrotated factor accounted for 32.4% of the total variance, well below the commonly used 50% threshold, suggesting that common method bias was unlikely to be a serious concern.

In Study 2, hypotheses were tested using analysis of variance and regression-based mediation and moderated mediation procedures. The factorial design allowed direct examination of the causal effects of regulatory clarity and enforcement posture on AI investment intentions, productivity expectations, deployment speed, and growth projections. By triangulating multilevel survey evidence with experimental manipulation, the two studies jointly reduce concerns related to common method variance and strengthen causal inference, providing convergent validity for the proposed model.

6. Results

6.1. Model specification

The hypothesized multilevel moderated mediation model was specified with firms (Level-1) nested within regions (Level-2). At Level-1, productivity gains were modeled as a function of AI-enabled innovation, while firm economic growth was modeled as a function of both productivity gains and AI-enabled innovation:

$$PG\_ij = \beta 0j + \beta 1j(AEI\_ij) + r\_ij$$

$$FEG\_ij = \gamma 0j + \gamma 1j(PG\_ij) + \gamma 2j(AEI\_ij) + e\_ij$$

At Level-2, intercepts and slopes were allowed to vary across regions, and the slope of AI-enabled innovation on productivity gains was modeled as a function of regional pro-innovation regulatory climate:

$$\beta 0j = \gamma 00 + u0j$$

$$\beta 1j = \gamma 10 + \gamma 11\left(PRC\_region\_j\right) + u1j$$

Firm-level predictors, including AI-enabled innovation, were group-mean centered prior to estimating cross-level interactions to isolate within-region effects. Intraclass correlation coefficients (ICCs) were computed to assess the appropriateness of a multilevel approach. The ICC(1) values were .18 for productivity gains and .21 for firm economic growth, indicating that 18–21% of the variance in these outcomes was attributable to between-region differences and justifying the use of hierarchical linear modeling.

6.1.1. Descriptive statistics and measurement model

Table 1 presents descriptive statistics, correlations, skewness, kurtosis, and ICCs for Study 1. Skewness and kurtosis values for all continuous variables were within acceptable thresholds, supporting the assumption of approximate normality and the use of parametric analyses. Correlations provided preliminary support for the hypothesized relationships. AI-enabled innovation was positively correlated with productivity gains ($r = .42$, $p < .01$) and firm economic growth ($r = .36$, $p < .01$), while productivity gains were strongly associated with firm economic growth ($r = .44$, $p < .01$). Pro-innovation regulatory climate also exhibited significant positive correlations with innovation, productivity, and growth at the regional level.

Before estimating structural relationships, the adequacy of the measurement model was assessed. As shown in Table 2, all constructs demonstrated strong internal consistency (Cronbach's α = .85–.91; composite reliability = .89–.93). Convergent validity was supported by average variance extracted (AVE) values ranging from .62 to .67, exceeding the .50 benchmark. Discriminant validity was established using the heterotrait–monotrait ratio (HTMT), with all values ranging between .68 and .72, well below the conservative threshold of .85. These results indicate that the measurement model exhibited satisfactory reliability and validity.

6.1.2. Multilevel regression and moderated mediation results (study 1)

Table 3 reports the multilevel regression results. Among control variables, firm size was positively associated with productivity gains ($\gamma = .09$, $p < .05$) and firm economic growth ($\gamma = .12$, $p < .05$). Firm age was positively related to firm economic growth ($\gamma = .11$, $p < .05$) but not to productivity gains, while IT budget intensity significantly predicted both productivity gains ($\gamma = .14$, $p < .01$) and firm economic growth ($\gamma = .16$, $p < .01$).

Turning to the focal predictors, AI-enabled innovation significantly predicted productivity gains ($\gamma = .28$, $p < .001$, 95% CI [.19, .37]), supporting H1. AI-enabled innovation also exerted a significant direct effect on firm economic growth ($\gamma = .15$, $p < .05$, 95% CI [.04, .26]), indicating that AI innovation contributes to growth beyond its indirect effects through productivity. Productivity gains were a strong predictor of firm economic growth ($\gamma = .39$, $p < .001$, 95% CI [.28, .50]), supporting H2.

At the regional level, pro-innovation regulatory climate had significant direct effects on productivity gains ($\gamma = .18$, $p < .05$) and firm economic growth ($\gamma = .10$, $p < .05$). Importantly, the cross-level interaction between AI-enabled innovation and regulatory climate was significant in predicting productivity gains ($\gamma = .16$, $p < .01$), indicating that the productivity benefits of AI-enabled innovation are amplified in supportive regulatory environments. Model fit statistics reported in Table 3 show that the full model fit the data significantly better than baseline models ($\Delta-2LL = 161.1$, $p < .001$), explaining 27% of the variance in productivity gains and 34% of the variance in firm economic growth. Significant random slope variance for AI-enabled innovation (variance = .10, $p < .05$) further indicates heterogeneity across regions.

Building on these findings, moderated mediation was examined. As shown in Table 4, the indirect effect of AI-enabled innovation on firm

Table 2
Measurement model results – Study 1.

Construct	Cronbach's α	CR	AVE	HTMT (max)
AI-enabled innovation	.91	.93	.67	.72
Productivity gains	.88	.91	.65	.70
Firm economic growth	.85	.89	.62	.68
Pro-innovation regulatory climate (PRC_region)	.87	.90	.64	.71

Table 1
Descriptive statistics, correlations, and ICCs – Study 1.

Variable	M	SD	Skewness	Kurtosis	1	2	3	4	5	6	7	8
1. Firm size (log)	3.15	.88	.21	−.47	—							
2. Firm age (years)	12.3	8.4	.64	−.12	.22**	—						
3. IT budget intensity (%)	6.1	2.4	.58	.31	.31**	.18*	—					
4. AI-enabled innovation	4.92	1.21	−.41	−.36	.27**	.19*	.33**	—				
5. Productivity gains	4.78	1.18	−.38	−.29	.23**	.21*	.29**	.42**	—			
6. Firm economic growth	5.01	1.32	−.27	−.18	.20**	.25**	.31**	.36**	.44**	—		
7. Pro-innovation regulatory climate (PRC_region)	5.15	.97	−.19	−.41	.18*	.15*	.22**	.29**	.31**	.27**	—	
8. Digital infrastructure index	6.42	1.04	.33	−.22	.16*	.14*	.19*	.22**	.21*	.20*	.34**	—
ICC(1) productivity gains	—	—	—	—	—	—	—	—	—	—	—	.18
ICC(1) firm economic growth	—	—	—	—	—	—	—	—	—	—	—	.21

Table 3
Multilevel regression, model fit, and random effects – Study 1.

Panel A: Fixed effects (standardized coefficients)						
Predictor	PG (γ)	FEG (γ)	SE	t	95% CI (Lower)	95% CI (Upper)
Firm size	.09*	.12*	.04	2.25	.01	.17
Firm age	.05 (ns)	.11*	.05	2.02	.01	.21
IT budget intensity	.14**	.16**	.05	2.90	.04	.24
AI-enabled innovation	.28***	.15*	.06	4.67	.19	.37
Productivity gains	—	.39***	.07	5.57	.28	.50
Pro-innovation regulatory climate (PRC_region)	.18*	.10*	.05	2.15	.02	.34
AEI × PRC_region	.16**	—	.06	2.73	.04	.28
Panel B: Model fit indices and variance explained						
Model	−2LL	AIC	BIC	Δ−2LL vs Null	R^2 (Pseudo)	—	—
Null (intercept only)	1589.2	1603.5	1631.2	—	—	—	—
Random intercept model	1492.6	1508.7	1538.9	96.6***	PG: .19, FEG: .22	—	—
Full model with interaction	1428.1	1447.2	1478.6	161.1***	PG: .27, FEG: .34	—	—
Panel C: Random effects variance components						
Random Effect	Variance	SE	p	—	—	
Intercept variance (PG)	.21	.07	<.01	—	—	
Slope variance (AEI → PG)	.10	.04	<.05	—	—	
Intercept variance (FEG)	.28	.09	<.01	—	—	
Residual variance	1.65	.12	<.001	—	—	

Table 4
Conditional indirect effects and simple slopes – Study 1.

Panel A: Conditional indirect effects of AEI on FEG via PG			
PRC_region Level	Indirect Effect	Boot SE	95% CI
Low (−1 SD)	.07 (ns)	.05	[-.02, .16]
Mean	.12*	.04	[.04, .21]
High (+1 SD)	.19**	.06	[.08, .31]
Panel B: Simple slopes of AEI predicting PG at levels of PRC_region			
PRC_region Level	Slope (γ)	SE	t
Low (−1 SD)	.18*	.07	2.57
Mean	.28***	.06	4.67
High (+1 SD)	.37***	.08	4.63

economic growth via productivity gains was not significant at low levels of regulatory support (indirect = .07, 95% CI [−.02, .16]). At mean regulatory support, the indirect effect became significant (indirect = .12, 95% CI [.04, .21]), and at high regulatory support the indirect effect was strongest (indirect = .19, 95% CI [.08, .31]). These results support H3, H4, and H5 and indicate that productivity gains mediate the innovation–growth relationship in a manner contingent on regulatory context.

6.1.3. *Experimental results (study 2)*

To complement the multilevel survey findings and strengthen causal inference, results from the experimental vignette study are reported next. Manipulation checks were examined prior to hypothesis testing. As shown in Table 7, participants in the clear regulatory condition reported significantly higher perceived clarity ($M = 5.9$, $SD = .9$) than those in the ambiguous condition ($M = 3.1$, $SD = 1.0$), $t(358) = 21.31$, $p < .001$, $d = 2.25$. Participants in the supportive enforcement condition perceived enforcement as significantly more supportive ($M = 5.6$, $SD = 1.0$) than those in the punitive condition ($M = 3.2$, $SD = 1.1$), $t(358) = 18.73$, $p < .001$, $d = 1.98$. These results confirm that both manipulations were successful.

Descriptive statistics for the experimental conditions are presented in Table 5. The clear–supportive condition yielded the highest intended AI investment ($M = 11.8$, $SD = 3.1$), expected productivity gains ($M = 10.2$, $SD = 2.8$), and expected firm growth ($M = 6.5$, $SD = 2.1$), along with the fastest intended deployment speed ($M = 4.3$ months, $SD = 1.2$). In contrast, the ambiguous–punitive condition produced the least favorable outcomes across all dependent variables.

Table 6 reports the results of two-way ANOVAs examining the effects of regulatory clarity, enforcement posture, and their interaction. For intended AI investment, clarity ($F = 34.80$, $p < .001$, $\eta^2p = .089$) and enforcement ($F = 28.41$, $p < .001$, $\eta^2p = .074$) exhibited significant main effects, along with a significant interaction ($F = 4.61$, $p = .033$, $\eta^2p = .013$). Similar patterns emerged for expected productivity gains (clarity: $F = 30.12$, $p < .001$, $\eta^2p = .078$; enforcement: $F = 24.23$, $p < .001$, $\eta^2p = .064$; interaction: $F = 5.06$, $p = .025$, $\eta^2p = .014$),

Table 6
Two-way ANOVAs for regulatory clarity (C), enforcement posture (E), and C × E interactions – Study 2.

Dependent variable	Effect	df	F	p	η^2p
AI Investment %	C	1, 356	34.80	<.001	.089
AI Investment %	E	1, 356	28.41	<.001	.074
AI Investment %	C × E	1, 356	4.61	.033	.013
Expected Productivity %	C	1, 356	30.12	<.001	.078
Expected Productivity %	E	1, 356	24.23	<.001	.064
Expected Productivity %	C × E	1, 356	5.06	.025	.014
Deployment Speed months	C	1, 356	22.47	<.001	.059
Deployment Speed months	E	1, 356	18.86	<.001	.050
Deployment Speed months	C × E	1, 356	3.92	.049	.011
Expected Growth %	C	1, 356	26.76	<.001	.070
Expected Growth %	E	1, 356	20.98	<.001	.056
Expected Growth %	C × E	1, 356	4.18	.042	.012

Table 5
Means and standard deviations by conditions for primary dependent variables – Study 2.

Condition	AI investment % (M, SD)	Expected productivity % (M, SD)	Deployment speed months (M, SD)	Expected growth % (M, SD)	n
Clear–supportive	11.8 (3.1)	10.2 (2.8)	4.3 (1.2)	6.5 (2.1)	90
Clear–punitive	9.6 (3.0)	8.7 (2.9)	5.1 (1.3)	5.4 (2.0)	90
Ambiguous–supportive	9.9 (3.2)	9.1 (3.0)	5.0 (1.4)	5.6 (2.1)	90
Ambiguous–punitive	8.1 (3.1)	7.5 (2.7)	5.8 (1.5)	4.2 (2.0)	90

Table 7
Bootstrap indirect effects on expected firm growth via expected productivity gains – Study 2.

Check	Level A (M, SD)	Level B (M, SD)	df	t	p	Cohen's d
Perceived regulatory clarity (clear vs ambiguous)	5.9 (.9)	3.1 (1.0)	358	21.31	<.001	2.25
Perceived enforcement posture (supportive vs punitive)	5.6 (1.0)	3.2 (1.1)	358	18.73	<.001	1.98

deployment speed, and expected firm growth. Across outcomes, effect sizes ranged from .050 to .089, indicating small-to-moderate effects.

Finally, mediation analyses were conducted to examine whether expected productivity gains transmitted the effects of regulatory conditions to growth expectations. As shown in Table 8, the indirect effect of regulatory clarity on expected firm growth via productivity was significant (b = .48, SE = .16, 95% CI [.19, .84]), as was the indirect effect of enforcement posture (b = .37, SE = .15, 95% CI [.08, .68]). The index of moderated mediation was also significant (b = .12, SE = .06, 95% CI [.01, .25]), indicating that the productivity-based mediation pathway from regulatory clarity to growth expectations was stronger under supportive enforcement conditions.

6.1.4. General discussion

This study sought to unpack how AI-enabled innovation contributes to firm-level economic growth through the mediating role of productivity gains and the boundary condition of regulatory climate. Drawing on a dual-study design, the findings provide consistent support for the proposed model, reinforcing the importance of AI-enabled innovation as a contributor to firm performance under varying institutional conditions. Table 9 summarizes the key results across both studies, enabling a direct comparison of mechanisms and effects across methodological approaches.

The empirical context of this study is situated within selected Asia-Pacific economies, including China, Malaysia, Singapore, and Thailand. While this regional focus allows examination of AI-enabled innovation within diverse but innovation-active environments, it also introduces potential country-level heterogeneity. Differences in regulatory maturity, enforcement consistency, and institutional norms across these contexts may influence how firms perceive regulatory support and translate AI capabilities into productivity gains and growth outcomes. Accordingly, the findings should be interpreted with appropriate caution, and future research is encouraged to explicitly examine cross-country variation in AI-related regulatory environments.

The findings suggest that AI-enabled innovation significantly enhances productivity gains, supporting prior research indicating that AI technologies improve efficiency, automate processes, and strengthen operational agility (Atienza-Barba et al., 2024; Chen et al., 2014). By enabling firms to streamline processes and allocate resources more effectively, AI functions as a strategic capability that can enhance organizational adaptability and performance (Chin et al., 2025; Cui, 2025). Importantly, this effect was observed across both realized firm outcomes and managerial expectations, extending prior work that has primarily focused on adoption or technical efficiency rather than productivity as a core economic mechanism (Fu et al., 2025).

Table 8
Bootstrap indirect effects on expected firm growth via expected productivity gains – Study 2.

Predictor	b	SE (boot)	95% CI (LL, UL)	z	p
Clarity (clear vs ambiguous)	.48	.16	.19, .84	2.98	.003
Enforcement (supportive vs punitive)	.37	.15	.08, .68	2.47	.013
Clarity × enforcement (index of moderated mediation)	.12	.06	.01, .25	2.02	.043

Table 9
Summary of empirical findings across Study 1 and Study 2.

Hypothesis/ Relationship	Study 1 (Survey, HLM)	Study 2 (Experiment)	Contribution vs prior literature
AI-enabled innovation → Productivity gains (H1)	Positive, significant (γ = .28, p < .001)	Supported via higher expected productivity under clear–supportive regulation	Extends AI–performance literature by showing productivity as a primary mechanism
Productivity gains → Firm growth (H2)	Positive, significant (γ = .39, p < .001)	Supported via productivity-mediated growth expectations	Clarifies internal efficiency as growth transmission channel
AI-enabled innovation → Firm growth (direct)	Positive, smaller effect (γ = .15, p < .05)	Reflected in growth expectations	Shows partial mediation rather than direct-only effects
Regulatory climate × AI → Productivity (H4)	Significant cross-level moderation	Significant interaction (clarity × enforcement)	Demonstrates institutional amplification of AI value (aligns with TIS policy studies)
Moderated mediation (H5)	Indirect effect stronger at high regulatory support	Indirect effect conditional on enforcement posture	Introduces conditional productivity pathway into AI-growth models

In turn, productivity gains were found to be a central mechanism through which AI-enabled innovation is associated with firm economic growth. This finding underscores that the economic value of AI is realized not through direct effects alone, but through internal efficiency improvements that enable firms to achieve scalability, cost advantages, and competitive positioning (Cimino et al., 2025a,b; Keskar, 2024; Parwatay, 2021). By clarifying this pathway, the study helps explain why prior research has reported inconsistent direct effects of AI adoption on growth, highlighting the critical role of productivity transformation in linking innovation to performance outcomes.

The results further indicate that these relationships are contingent on the regulatory environment. Specifically, pro-innovation regulatory climates appear to amplify the productivity benefits of AI-enabled innovation, supporting institutional perspectives that emphasize the role of external policy contexts in shaping firm-level outcomes (Rudko et al., 2025). In environments characterized by regulatory clarity and supportive enforcement, firms are better able to experiment with and scale AI solutions, whereas ambiguous or restrictive conditions may hinder implementation and reduce returns (Chan et al., 2024; Gama & Magistretti, 2025).

Building on this, the moderated mediation findings show that the indirect effect of AI-enabled innovation on firm growth via productivity gains is stronger in supportive regulatory contexts. This highlights that AI-driven growth is not universally linear but depends on the alignment between internal capabilities and external institutional support (Govindan, 2022; Nikolova & Angrisani, 2025). By integrating multilevel and experimental evidence, the study demonstrates how regulatory signals shape both actual firm behavior and managerial expectations regarding AI investment and performance outcomes.

At the same time, the findings should be interpreted with attention to regional specificity. The data were collected from firms operating in the Asia-Pacific region, where regulatory regimes and institutional approaches to AI governance may differ from those in other regions. These contextual differences may influence how firms perceive regulatory support and translate AI capabilities into performance outcomes.

Accordingly, caution is warranted when generalizing the findings, and future research should test the model across diverse institutional settings.

Overall, this study advances understanding of AI-enabled innovation by positioning productivity gains and regulatory alignment as central mechanisms shaping firm growth. By synthesizing evidence across complementary methodological approaches, the findings provide a more integrated explanation of when and how AI investments generate economic value.

6.2. Theoretical implications

This study contributes to the literature on innovation, firm performance, and institutional dynamics by advancing and empirically validating a moderated mediation model of AI-enabled innovation and firm growth. In doing so, it offers a more precise understanding of how and under what conditions AI contributes to economic outcomes. First, the study adds conceptual clarity to the innovation–performance nexus by demonstrating that AI-enabled innovation contributes to economic growth primarily through productivity enhancement rather than direct effects. While prior research has recognized AI's potential to elevate firm capabilities (Aldoseri et al., 2024; Burström et al., 2021), our findings specify the internal transformation mechanism—centered on efficiency gains—through which these benefits are realized. This aligns with emerging perspectives emphasizing process-level changes as the foundation of sustainable growth (Füller et al., 2024; Tekic & Füller, 2023), and advances existing research that often conflates adoption, capability development, and performance outcomes into a single process.

Second, the study extends dynamic capabilities theory by conceptualizing AI-enabled innovation as a strategic resource that enhances firms' ability to reconfigure operations and adapt to changing environments. By linking AI-driven capability development to productivity gains and growth, the findings position AI not merely as a technological upgrade but as a mechanism of organizational renewal and agility (Ateeq et al., 2025; Atienza-Barba et al., 2024). Importantly, the results further show that these capabilities yield stronger performance outcomes when supported by favorable institutional conditions, complementing recent work on the contingent nature of capability-based advantages (Renfei & Zhongwen, 2025).

Third, the findings refine institutional theory by demonstrating that regulatory environments function as active enablers of innovation outcomes rather than passive constraints. While prior research has examined institutional influences on innovation (Erdmann & Toro-Dupouy, 2025), this study highlights how perceived regulatory clarity and supportiveness strengthen the translation of innovation into productivity gains. This contribution positions regulatory climate as a boundary condition that amplifies firm-level performance effects, reinforcing the importance of contextual alignment in innovation processes (Chan et al., 2024; Memmert et al., 2023).

Fourth, the study contributes to AI management research by empirically linking AI-enabled innovation to tangible firm-level outcomes within a theoretically grounded framework. Moving beyond adoption-focused perspectives, the findings situate AI within performance and capability logics, supporting the view that AI should be understood as an embedded organizational capability rather than a standalone technological artifact (Sandeep et al., 2025; Teece, 2014). This perspective is consistent with emerging work on how AI-related knowledge and institutional signals shape managerial expectations and strategic behavior (Nikolova & Angrisani, 2025).

Finally, the dual-study design enhances theoretical robustness by demonstrating convergence across multilevel field data and experimental evidence. By capturing both realized firm behavior and controlled managerial decision-making, the study strengthens confidence in the proposed productivity-centered pathway linking AI-enabled innovation to firm growth. This multi-method approach provides a foundation for future research examining AI-driven transformation across levels and contexts.

6.3. Practical implications

The findings of this study offer several practical insights for organizational leaders, policymakers, and technology strategists aiming to leverage AI as a catalyst for economic performance. At the same time, it is important to acknowledge that the practical recommendations derived from this study are inherently context-dependent. The feasibility and effectiveness of AI-enabled innovation strategies may vary considerably across firm sizes, sectors, and institutional environments, particularly between advanced and emerging economies. Accordingly, the following implications should be interpreted as conditional guidance rather than universally applicable prescriptions.

First, for business leaders and executives, the study underscores that AI adoption alone is insufficient to guarantee economic gains. Instead, firms must prioritize AI-enabled innovation—that is, embedding AI technologies into the design of new products, processes, and service models. To realize measurable productivity improvements, managers should move beyond pilot experiments and short-term deployments, and instead foster organization-wide AI integration that aligns with strategic goals and internal workflows. Investments in AI should therefore be accompanied by complementary initiatives such as employee upskilling, cross-functional data integration, and process reengineering.

However, such organization-wide integration presupposes the availability of financial resources, technical expertise, and absorptive capacity. Smaller firms and organizations operating under resource constraints may need to adopt a more incremental approach, focusing on targeted AI applications with clear productivity payoffs rather than large-scale transformation efforts.

Second, the central role of productivity as a mediating mechanism suggests that organizations need to treat internal efficiency not merely as an operational concern but as a strategic enabler of growth. Firms should establish KPIs and monitoring systems that track how AI-driven innovations contribute to reductions in process latency, error rates, and resource wastage. Building a culture of continuous improvement—where AI tools are used to iteratively refine operations—can help bridge the gap between innovation investment and financial returns. This emphasis on productivity metrics also serves as a safeguard against symbolic or premature AI adoption, ensuring that technological investments are evaluated based on tangible efficiency gains rather than reputational or mimetic pressures.

Third, for governments and regulators, the moderating effect of regulatory climate reveals that supportive institutional environments significantly amplify the benefits of AI innovation. Policymakers should therefore aim to craft clear, stable, and innovation-friendly regulations that reduce legal uncertainty for firms experimenting with AI. This includes setting ethical boundaries and data protection standards, while simultaneously enabling sandbox environments, tax incentives, or public–private AI pilot programs. Regions that prioritize regulatory clarity and enforcement consistency are likely to attract more AI-driven innovation and associated economic activity.

Nevertheless, regulatory reform is often constrained by political, administrative, and societal considerations. Policymakers should therefore view regulatory clarity as an evolving process rather than an immediate outcome, balancing innovation facilitation with risk management, ethical oversight, and public trust. Overregulation or overly rigid compliance frameworks may inadvertently suppress experimentation or generate new forms of organizational risk.

Fourth, firms operating in emerging or less supportive regulatory environments should proactively engage in institutional navigation strategies. These may include building alliances with government stakeholders, participating in policy dialogues, or creating internal governance structures that align AI initiatives with compliance mandates. Such anticipatory efforts can mitigate the friction caused by institutional ambiguity and protect innovation investments from regulatory setbacks. At the same time, the capacity to engage in such strategies is unevenly distributed. Smaller firms may lack the influence, visibility, or resources required for active institutional engagement, underscoring the importance of intermediary actors—such as industry associations or innovation hubs—in representing collective interests and lowering participation barriers.

Finally, for industry associations and ecosystem builders, the results suggest a need to facilitate cross-sector knowledge sharing, benchmarking, and diffusion of best practices in AI implementation. By building networks that connect firms, researchers, and policy experts, these actors can foster collective learning and accelerate the maturity of AI-enabled innovation ecosystems. Such ecosystem-level coordination is particularly critical in regions characterized by regulatory fragmentation, where shared standards and collective learning can partially compensate for institutional uncertainty.

In essence, this study provides a strategic roadmap for turning AI investments into sustainable performance gains—one that depends not only on technological capability but also on operational readiness and regulatory alignment.

7. Limitations

Despite its contributions, this study is subject to several limitations that offer fertile ground for future research. First, while we employed a dual-study design to enhance robustness and generalizability, both datasets were cross-sectional in nature, limiting our ability to draw definitive causal inferences. Future research could adopt longitudinal or experimental designs to better capture the dynamic interplay between AI-enabled innovation, productivity trajectories, and economic outcomes over time.

Second, although we conceptualized productivity gains as a key mediating mechanism, our operationalization primarily captured internal efficiency metrics. Future studies may benefit from incorporating a more holistic view of productivity that includes employee well-being, creativity, or customer satisfaction—dimensions that are increasingly relevant in AI-transformed workplaces.

Third, our measure of regulatory climate focused on firms' subjective perceptions of regulatory clarity and enforcement. While this aligns with prior institutional theory research, incorporating objective regulatory indices or triangulating with country-level policy datasets (e.g., OECD AI indicators) could yield a more nuanced understanding of how formal institutions shape innovation outcomes.

Fourth, although the survey data were collected from firms operating across multiple regions, the empirical context of Study 1 is situated within the Asia-Pacific region. As regulatory cultures, institutional norms, and approaches to AI governance vary internationally, caution is warranted when extrapolating the findings to regions with markedly different institutional architectures. Future research is encouraged to test the proposed model across diverse geographical settings and regulatory regimes to assess its cross-contextual robustness.

Fifth, this study was conducted in industry contexts where AI adoption is relatively mature. Generalizability to sectors with lower technological readiness or in early stages of digital transformation may be limited. Future research should explore boundary conditions across sectoral maturity levels, firm sizes (e.g., SMEs), and digital competencies to test the scalability of our model.

Sixth, while we positioned AI-enabled innovation as a firm-level capability, we did not explore micro-foundations such as leadership styles, employee readiness, or AI governance structures. Investigating how individual- and team-level factors interact with firm strategies may offer deeper insights into the capability-building process.

Seventh, while Study 2 strengthens causal inference through experimental manipulation, the vignette-based design necessarily simplifies complex real-world regulatory environments into stylized scenarios. Actual AI adoption decisions may be shaped by additional considerations—such as budget constraints, organizational politics, legacy systems, and ethical deliberations—that cannot be fully captured in controlled experimental settings.

Last but not the least, we grounded our framework in dynamic capabilities and institutional theory, other theoretical perspectives—such as socio-technical systems theory, absorptive capacity, or attention-based view—could be used to expand or reinterpret the relationships proposed in our model. Future scholars are encouraged to explore alternative theoretical lenses or test competing models to further enrich the conceptual understanding of AI's organizational value.

8. Conclusion

8.1. Theoretical contributions

This study advances understanding of AI-enabled innovation by demonstrating that its impact on firm economic growth is primarily realized through productivity gains, rather than direct effects alone. By integrating dynamic capabilities and institutional theory within a moderated mediation framework, the findings offer a more nuanced explanation of how internal efficiency transformation and external regulatory conditions jointly shape AI-driven performance outcomes.

8.2. Practical implications

For practitioners, the results highlight that AI adoption should ideally be embedded into core operational processes to generate meaningful productivity improvements and growth. Firms should therefore focus on aligning AI initiatives with organizational workflows and capability development. At the same time, the findings underscore the importance of regulatory alignment, as supportive policy environments significantly enhance the effectiveness of AI-enabled innovation. For policymakers, this suggests the need for clear, stable, and innovation-friendly regulatory frameworks that reduce uncertainty while enabling responsible experimentation.

8.3. Future research directions

This study opens several avenues for future research. Longitudinal investigations could provide deeper insight into how AI-enabled innovation and productivity evolve over time. Future studies may also explore micro-level mechanisms, such as leadership, governance, and employee readiness, that influence the effectiveness of AI adoption. Additionally, cross-regional comparisons would help clarify how different institutional environments shape the relationship between AI-enabled innovation, productivity gains, and firm growth.

CRediT authorship contribution statement

Zhuzhou Liu: Writing – review & editing, Writing – original draft, Validation, Software, Project administration, Investigation, Formal analysis, Conceptualization. **Zhe Weng:** Writing – review & editing, Writing – original draft, Validation, Software, Project administration, Investigation, Funding acquisition, Data curation. **Shichao Yu:** Writing – review & editing, Writing – original draft, Validation, Software, Project administration, Investigation, Formal analysis, Conceptualization.

Conflict of interests

The author declares that there is no conflict of interest regarding the publication of this article.

Data availability

Data will be made available on request.

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